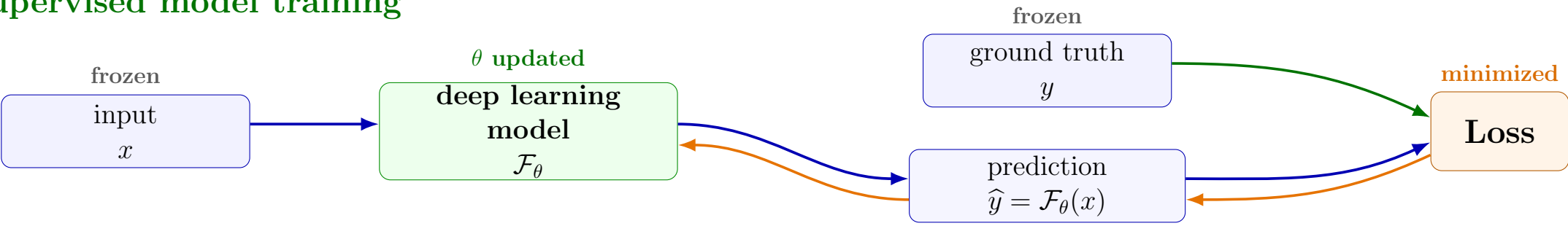


Supervised Model Training and Adversarial Attack in the Usual Context

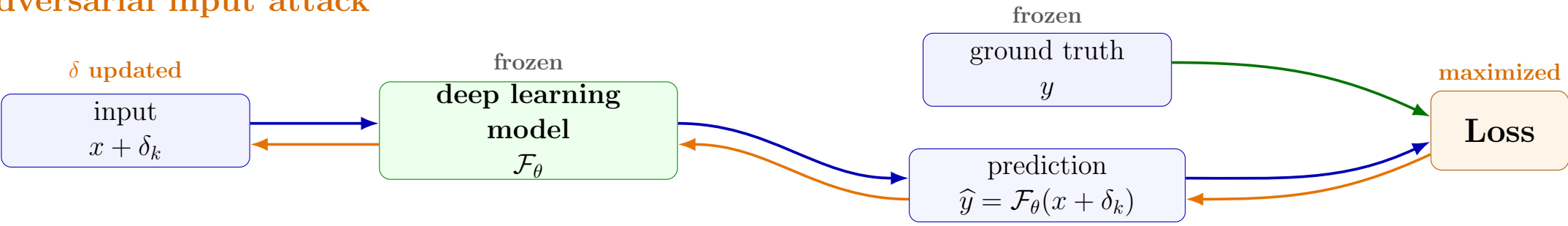
→ model forward → ground truth forward → model backward

Supervised model training



$$\mathcal{L}_{\text{train}}(\theta; x, y) = \frac{1}{2} \|\mathcal{F}_\theta(x) - y\|_2^2, \quad \nabla_\theta \mathcal{L}_{\text{train}} = D_\theta \mathcal{F}_\theta(x)^* (\mathcal{F}_\theta(x) - y), \quad \theta_{t+1} = \theta_t - \alpha \nabla_\theta \mathcal{L}_{\text{train}}(\theta_t; x, y).$$

Adversarial input attack



$$x_k = x + \delta_k, \quad \mathcal{L}_{\text{attack}}(\delta_k; x, y) = \frac{1}{2} \|\mathcal{F}_\theta(x_k) - y\|_2^2, \quad \nabla_\delta \mathcal{L}_{\text{attack}} = D_x \mathcal{F}_\theta(x_k)^* (\mathcal{F}_\theta(x_k) - y), \quad \delta_{k+1} = \Pi_\Delta(\delta_k + \eta \nabla_\delta \mathcal{L}_{\text{attack}}).$$